

Credit Card Fraud Detection Using Deep Learning

An MLP + Autoencoder Stacking Ensemble Approach

ETA3_AI — AI Training Program (Huawei - NTI)

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Fraud is rare enough to make accuracy misleading

DATASET

283,726

transactions after duplicate removal

473

fraud · 0.167%

Accuracy alone is misleading. Predicting legitimate for almost every transaction can look excellent while missing the fraud cases that matter.

EVALUATION LENSES

Precision · Recall · F1

ROC-AUC · PR-AUC · False positives

Thirty engineered features make the input reproducible

The same feature order is used in training and deployment.

1

V1-V28

28 anonymized PCA features preserve the transformed transaction signal.

2

Amount

$\text{Amount_log} = \log_{1p}(\text{Amount})$,
reducing extreme-value skew.

3

Hour

$\text{Hour} = (\text{Time} // 3600) \% 24$

Raw Time and Amount are removed;
Amount is transformed into Amount_log
using log1p.

Data split and leakage control

Stratified split

Train: 181,584

Validation: 45,396

Test: 56,746

Scale

StandardScaler is fitted only on training data, then applied to validation and test.

Balance

SMOTE is applied only to training data. Validation and test retain the real class distribution.

Overall system architecture

Two complementary signals are calibrated and combined into one final decision.

INPUT

30

engineered features

MLP

Fraud probability

Autoencoder

Reconstruction error

META-MODEL

Meta-scaler → Logistic
Regression → threshold
0.9885

30 features → MLP + Autoencoder → meta-features → ensemble score → Fraud / Legitimate

The MLP learns a non-linear fraud boundary

ARCHITECTURE

30 → 128 → 64 → 32 → 16 → 1

ReLU hidden layers · sigmoid output

Dropout: 0.40 → 0.30 → 0.20

Training controls

SMOTE-balanced training

EarlyStopping on validation PR-AUC

ReduceLROnPlateau

Threshold selected by maximizing
Recall on validation — not by
maximizing F1.

The MLP catches the most fraud, at a high false-alarm cost

MLP TEST RESULTS · THRESHOLD 0.01

Accuracy	99.85%
Precision	53.42%
Recall	82.11%
F1	64.73%
ROC-AUC	95.22%
PR-AUC	79.86%
False positives	68

WHY IT MATTERS

82.11%

highest Recall

The MLP also has the highest PR-AUC, but its low threshold creates more false alarms.

The Autoencoder is precise, with fewer false alarms

AUTOENCODER TEST RESULTS · THRESHOLD 6.6521

Threshold selected on the validation set by maximizing F2-score

Accuracy	99.92%
Precision	80.23%
Recall	72.63%
F1	76.24%
ROC-AUC	94.05%
PR-AUC	65.01%
False positives	17

READOUT

69 / 95

true positives

Precision is stronger than the MLP, with fewer false alarms.

The Autoencoder learns normality, then flags deviation

It learns the structure of normal transactions and scores how far an input deviates.

ARCHITECTURE

30 → 32 → 16 → 8 → 16 → 32 → 30

Normal transactions only · RobustScaler

Gaussian noise · L2 regularization

SIGNAL

Weighted reconstruction error flags anomalous behavior.

Fraud-correlated features include V17, V14, V12, V10.

Stacking turns two signals into one decision

The meta-model sees both the MLP probability and the Autoencoder reconstruction error.

META-FEATURES

MLP fraud probability

Autoencoder reconstruction error

CALIBRATION

Meta-scaler: StandardScaler

Meta-model: Logistic Regression

class_weight="balanced"

FINAL DECISION

0.7446

ensemble threshold

Fraud / Legitimate

The ensemble improves the operating balance

ENSEMBLE TEST RESULTS · THRESHOLD 0.7446

Accuracy	99.94%
Precision	86.90%
Recall	76.84%
F1	81.56%
ROC-AUC	95.72%
False positives	11

READOUT

73 / 95

true positives

Best overall balance and fewest false positives on the held-out test set.

Model comparison

The ensemble is the balance choice, not a universal winner.

Model	Precision	Recall	F1	ROC-AUC	PR-AUC	FP
MLP	53.42%	82.11%	64.73%	95.22%	79.86%	68
Autoencoder	80.23%	72.63%	76.24%	94.05%	65.01%	17
Ensemble	86.90%	76.84%	81.56%	95.72%	72.42%	11

Ensemble: best overall balance and fewest false positives. MLP: highest Recall and PR-AUC.

Streamlit deployment

Inference mirrors the training-time pipeline without shipping the training dataset.

INFERENCE FLOW

User inputs transaction features

- same preprocessing
- MLP probability
- Autoencoder reconstruction error
- ensemble meta-model
- final threshold
- **FRAUD / LEGITIMATE**

DEPLOYMENT RULES

Saved artifacts only.

Does not load the training dataset.

Does not retrain models.

FINAL TAKEAWAY

Deep learning can detect highly imbalanced fraud patterns.

MLP provides strong fraud sensitivity.

Autoencoder detects deviations from normal behavior.

Stacking combines both signals.

Ensemble achieved the best overall balance on the held-out test set.

Final deployment is available through Streamlit.